DATA WAREHOUSE VS DATA LAKE VS DATA MART

Data Architecture - Quick Revision Notes

1 Core Definitions at a Glance

Data Warehouse (DWH): A central, structured data repository optimized for analytical processing (OLAP). Data is cleaned, transformed, and organized into schemas before loading (Schema-on-Write).

Data Lake: A central repository that stores massive volumes of raw, uncurated data in its native format (structured, semi-structured, unstructured). Schema is defined upon reading (Schema-on-Read).

Data Mart: A subset or focused slice of a Data Warehouse tailored specifically to the needs of a single business unit, department, or domain (e.g., Sales, Finance, HR).

2 Comprehensive Comparison Table

Feature	Data Warehouse	Data Lake	Data Mart
Data Structure	Structured (Relational / Star Schema)	Structured, Semi-structured, Unstructured	Structured (Domain-specific schemas)
Schema Strategy	Schema-on-Write (ETL)	Schema-on-Read (ELT)	Schema-on-Write (ETL)
Primary Purpose	Enterprise BI, SQL reporting, historical trends	Data science, ML, big data stream processing	Departmental reporting & targeted analytics
Primary Users	BI Analysts, Enterprise Reporting Teams	Data Engineers, Data Scientists	Business Unit Analysts (Finance, Marketing)
Cost & Storage	Higher storage cost, highly optimized compute	Very low storage cost (e.g., S3, ADLS Blob)	Moderate cost, smaller footprint
Query Speed	Fast to extremely fast for structured SQL	Variable (slower for raw, fast for Parquet/Iceberg)	Ultra-fast for targeted department queries

3 Key Architectural Paradigms: ETL vs ELT

- **Data Warehouse (ETL):** Extract from source → Transform in staging → Load clean schema into DWH.
- **Data Lake (ELT):** Extract from source → Load raw data immediately into storage → Transform on-demand when querying.

Key Insight: Data Lakes enable storage of "raw history" so future data science models can extract features that were never anticipated during initial ingestion.

DEEP DIVE & MODERN LAKEHOUSE PARADIGM

Architecture Implementation & Modern Data Stack Integration

4 Data Mart Approaches: Dependent vs Independent

Dependent Data Mart: Populated directly from an enterprise Data Warehouse. Ensures single source of truth, consistent metrics, and standardized governance across departments.

Independent Data Mart: Built directly from raw operational systems, bypassing a central warehouse. Faster to set up initially, but creates data silos and inconsistent metric definitions across teams.

Best Practice Rule: Always build *dependent* data marts to maintain a single source of truth across the enterprise.

5 The Evolution: What is a Data Lakehouse?

A **Lakehouse** combines the cheap, scalable object storage of a Data Lake with the ACID transaction capabilities, schema enforcement, and query performance of a Data Warehouse.

Technology Category	Examples	Primary Characteristics
Data Warehouse	Snowflake, Google BigQuery, AWS Redshift	Managed storage/compute, SQL optimized, high performance
Data Lake Storage	AWS S3, Azure Data Lake (ADLS), GCS	Object storage, cheap, unlimited scaling, decoupled compute
Lakehouse Table Formats	Apache Iceberg, Delta Lake, Apache Hudi	ACID transactions, time travel, schema evolution over object storage

6 When to Use Which Architecture?

• **Choose a Data Warehouse when:** You need reliable BI dashboards, structured historical reporting, strict data governance, and high-concurrency SQL queries.

• **Choose a Data Lake when:** You ingest high-velocity logs, sensor/IoT data, images, JSON, or require unstructured data for Machine Learning models.

• **Choose a Data Mart when:** A specific team (e.g., Marketing) needs isolated, highly performant access to their domain metrics without querying the entire DWH.

7 High-Yield Technical Interview Questions

Q1: What is a "Data Swamp" and how does it relate to a Data Lake?
Answer: A Data Swamp occurs when a Data Lake lacks proper governance, metadata management, schema validation, and access controls. It turns into an unorganized dumping ground where data cannot be audited, discovered, or reliably used for analytics.

Q2: Explain Schema-on-Write vs Schema-on-Read.
Answer: Schema-on-Write (Data Warehouse) mandates that structure, datatypes, and constraints are defined before inserting data into tables (via ETL). Schema-on-Read (Data Lake) accepts raw data as-is (via ELT), and the structure is applied dynamically when a query engine (like Spark or Trino) reads the files.

Q3: Why would a company need both a Data Lake and a Data Warehouse?
Answer: A dual-tier setup uses the Data Lake as an landing zone for raw, multi-format data and long-term ML experimentation, while refined data is transformed and loaded into the Data Warehouse for business analytics and executive KPI reporting.

Q4: What is the main drawback of building independent Data Marts?
Answer: Independent Data Marts pull data directly from source systems without a central warehouse. This leads to duplicate ETL pipelines, logic fragmentation, and conflicting business metrics (e.g., Sales and Finance calculating "Revenue" differently).

Q5: How do modern open table formats like Apache Iceberg bridge the gap between Data Lakes and Data Warehouses?
Answer: Apache Iceberg brings warehouse features—such as ACID transactions, hidden partitioning, time travel, and schema evolution—directly onto raw files (Parquet/ORC) stored in cheap cloud object storage, giving rise to Lakehouse architecture.